



SILVER OAK UNIVERSITY
School of Technology, Design And Computer Application
Silver Oak College of Computer Application
Master of Science Data Science
Course Name: Data Analytics Using Tools
Course Code: DSM5012C
Semester: 2nd

Prerequisite: Fundamental knowledge of data management, statistical concepts, and prior experience with spreadsheets, basic data visualization techniques, and familiarity with analytical reasoning and problem-solving approaches.

Course Objective: To develop students' competencies in data analytics by enabling them to acquire, process, explore, visualize, and interpret data, implement statistical and analytical techniques, design and apply predictive and prescriptive analytics workflows, and generate actionable insights for informed, data-driven decision-making in academic and professional contexts.

Teaching Scheme:

Teaching Scheme				
L	T	P	Contact Hours	Credit
4	0	0	4	4

Content:

Unit No.	Course Contents	Teaching Hours	Weightage %
1	Analytics Foundations Introduction to data analytics and analytical thinking, evolution and scope of data analytics in industry, data analytics vs data science, machine learning and artificial intelligence overview, types of analytics including descriptive and diagnostic analytics, predictive and prescriptive analytics, structured data concepts, semi-structured and unstructured data concepts, data sources including databases and transactional systems, APIs, IoT and log data, CRISP-DM methodology and analytics lifecycle, data warehousing concepts, data lakes and data mining fundamentals, ethics, privacy and data governance	14	25
2	Data Processing and Preparation Analytics workflow and environment overview, numerical and categorical data representation, tabular data structures and schema understanding, data ingestion concepts and formats, data profiling and quality assessment, handling missing values, outlier detection and treatment, duplicate detection and data consistency, data normalization and scaling concepts, encoding and feature transformation, feature creation and selection concepts, univariate exploratory data analysis, bivariate and multivariate analysis, correlation analysis concepts	14	25

3	Visualization and Statistical Analysis Importance of data visualization in analytics, principles of effective visualization, chart selection for categorical data, chart selection for numerical data, distribution analysis concepts, measures of central tendency, measures of dispersion, probability distribution overview, introduction to inferential statistics, hypothesis testing concepts, t-test and ANOVA concepts, chi-square test concepts, correlation methods and interpretation, regression analysis concepts	14	25
4	Advanced Analytics Tools and Applications Advanced data preparation techniques, analytical workflow automation concepts, introduction to predictive analytics, supervised analytics concepts, unsupervised analytics concepts, classification techniques overview, clustering techniques overview, association rule mining concepts, model performance evaluation metrics, confusion matrix and ROC concepts, cross-validation concepts, end-to-end analytics workflow integration, retail analytics case study, customer churn and market basket analysis case study	14	25

Course Outcome:

Sr.No.	CO-Statement	Unit
CO-1	Identify fundamental concepts of data analytics, types of analytics, data types, CRISP-DM methodology, and ethical governance	1
CO-2	Perform data processing, cleaning, transformation, feature engineering, and exploratory data analysis	2
CO-3	Use statistical measures, correlation, regression, and visualization techniques to derive insights from data	3
CO-4	Integrate advanced analytics workflows, predictive and unsupervised modeling, and industry-oriented case studies to Analyze	4

Teaching & Learning Methodology:

1. Lecture-Based Learning
2. Hands-On Labs and Practical Sessions
3. Project-Based Learning
4. Group Activities and Collaborative Learning
5. Active Learning Strategies

Major Equipment:

1. Computers/Laptops with internet connectivity
2. Projector and display system for demonstrations

